

Chapter 2

Descriptive Statistics

GLOBAL
EDITION



Elementary Statistics

Picturing the World

EIGHTH EDITION

Ron Larson



Chapter Outline

- 2.1 Frequency Distributions and Their Graphs
- 2.2 More Graphs and Displays
- 2.3 Measures of Central Tendency
- 2.4 Measures of Variation
- 2.5 Measures of Position

Section 2.4

Measures of Variation

Section 2.4 Objectives

- How to find the range of a data set
- How to find the variance and standard deviation of a population and of a sample
- How to use the Empirical Rule and Chebychev's Theorem to interpret standard deviation
- How to approximate the sample standard deviation for grouped data
- How to use the coefficient of variation to compare variation in different data sets

Range

Range

- The difference between the maximum and minimum data entries in the set.
- The data must be quantitative.
- $\text{Range} = (\text{Max. data entry}) - (\text{Min. data entry})$

Example: Finding the Range

Two corporations each hired 10 graduates. The starting salaries for each graduate are shown. Find the range of the starting salaries for Corporation A.

Starting Salaries for Corporation A (in thousands of dollars)

Salary	51	48	49	55	57	51	54	51	47	52
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Starting Salaries for Corporation B (in thousands of dollars)

Salary	50	33	51	60	59	42	51	39	62	68
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Solution: Finding the Range

- Ordering the data helps to find the least and greatest salaries.

47 48 49 51 51 51 52 54 55 57

minimum maximum

$$\begin{aligned}\text{Range} &= (\text{Max. salary}) - (\text{Min. salary}) \\ &= 57 - 47 = 10\end{aligned}$$

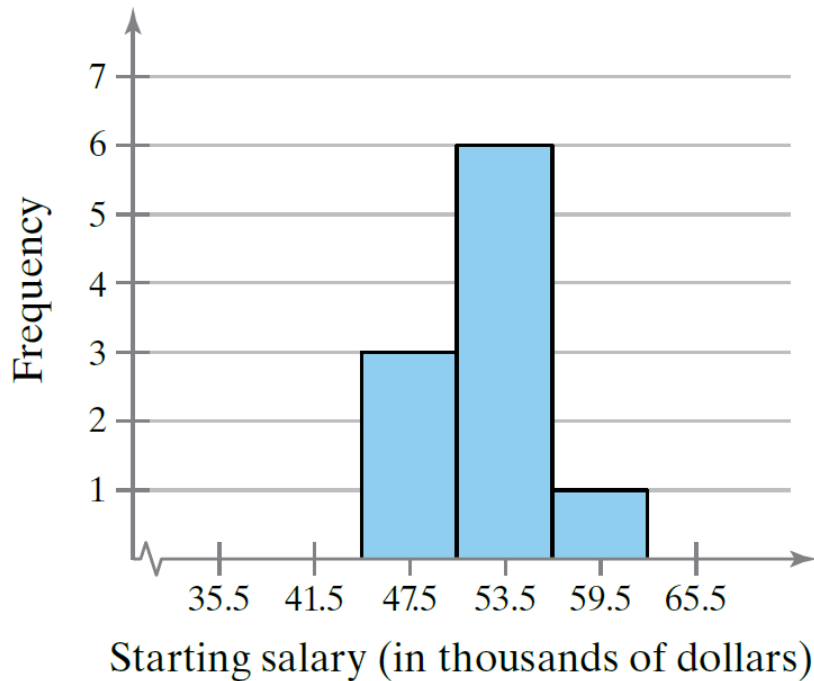
The range of starting salaries for Corporation A is 10, or \$10,000.

Variation

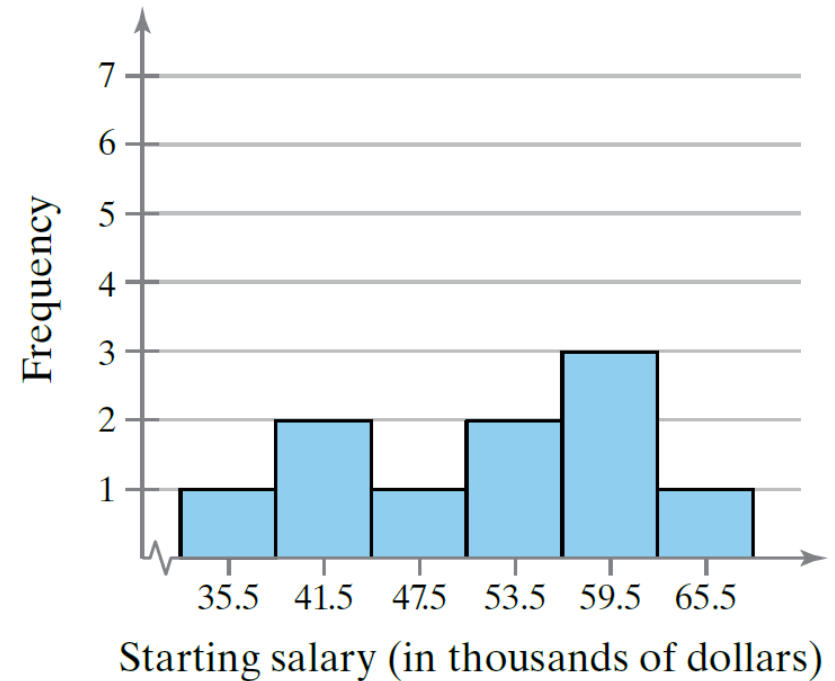
- Both data sets in the last example have a mean of 51.5, or \$51,500, a median of 51, or \$51,000, and a mode of 51, or \$51,000. And yet the two sets differ significantly.
- The difference is that the entries in the second set have greater variation. As you can see in the figures on the next slide, the starting salaries for Corporation B are more spread out than those for Corporation A.

Variation

Corporation A



Corporation B



Deviation, Variance, and Standard Deviation

Deviation

- The difference between the data entry, x , and the mean of the data set.
- Population data set:
 - Deviation of $x = x - \mu$
- Sample data set:
 - Deviation of $x = x - \bar{x}$

Deviation, Variance, and Standard Deviation

Population Variance

$$\sigma^2 = \frac{\Sigma(x - \mu)^2}{N}$$

Population Standard Deviation

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\Sigma(x - \mu)^2}{N}}$$

Deviation, Variance, and Standard Deviation

Observations About Standard Deviation

- The standard deviation measures the variation of the data set about the mean and has the same units of measure as the data set.
- The standard deviation is always greater than or equal to 0. When $\sigma = 0$, the data set has no variation and all entries have the same value.
- As the entries get farther from the mean (that is, more spread out), the value of σ increases.

Finding Population Variance & Standard Deviation

In Words

1. Find the mean of the population data set.
2. Find deviation of each entry.
3. Square each deviation.
4. Add to get the sum of squares.

In Symbols

$$\mu = \frac{\Sigma x}{N}$$

$$x - \mu$$

$$(x - \mu)^2$$

$$SS_x = \Sigma(x - \mu)^2$$

Finding the Population Variance & Standard Deviation

In Words

5. Divide by N to get the **population variance**.
6. Find the square root to get the **population standard deviation**.

In Symbols

$$\sigma^2 = \frac{\Sigma(x - \mu)^2}{N}$$

$$\sigma = \sqrt{\frac{\Sigma(x - \mu)^2}{N}}$$

Example: Finding Population Variance and Standard Deviation

Find the population variance and standard deviation of the starting salaries for Corporation A listed in the first Example.

For this data set, $N = 10$, $\sum x = 515$.

The mean is $\mu = \frac{515}{10} = 51.5$.

Solution: Finding Population Standard Deviation

- Determine the deviation for each data entry.

Salary (\$1000s), x	Deviation: $x - \mu$
51	$51 - 51.5 = -0.5$
48	$48 - 51.5 = -3.5$
49	$49 - 51.5 = -2.5$
55	$55 - 51.5 = 3.5$
57	$57 - 51.5 = 5.5$
51	$51 - 51.5 = -0.5$
54	$54 - 51.5 = 2.5$
51	$51 - 51.5 = -0.5$
47	$47 - 51.5 = -4.5$
52	$52 - 51.5 = 0.5$

$$\Sigma x = 515$$

$$\Sigma(x - \mu) = 0$$

Solution: Finding Population Standard Deviation

- Determine SS_x

Salary, x	Deviation: $x - \mu$	Squares: $(x - \mu)^2$
51	$51 - 51.5 = -0.5$	$(-0.5)^2 = 0.25$
48	$48 - 51.5 = -3.5$	$(-3.5)^2 = 12.25$
49	$49 - 51.5 = -2.5$	$(-2.5)^2 = 6.25$
55	$55 - 51.5 = 3.5$	$(3.5)^2 = 12.25$
57	$57 - 51.5 = 5.5$	$(5.5)^2 = 30.25$
51	$51 - 51.5 = -0.5$	$(-0.5)^2 = 0.25$
54	$54 - 51.5 = 2.5$	$(2.5)^2 = 6.25$
51	$51 - 51.5 = -0.5$	$(-0.5)^2 = 0.25$
47	$47 - 51.5 = -4.5$	$(-4.5)^2 = 20.25$
52	$52 - 51.5 = 0.5$	$(0.5)^2 = 0.25$

$$\Sigma(x - \mu) = 0$$

$$SS_x = 88.5$$

Solution: Finding Population Standard Deviation

Population Variance

- $$\sigma^2 = \frac{\Sigma(x - \mu)^2}{N} = \frac{88.5}{10} \approx 8.9$$

Population Standard Deviation

- $$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{88.5}{10}} \approx 3.0$$

The population variance is about 8.9, and the population standard deviation is about 3.0, or \$3,000.

Deviation, Variance, and Standard Deviation

Sample Variance

$$s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1}$$

Sample Standard Deviation

$$s = \sqrt{s^2} = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}$$

Finding the Sample Variance & Standard Deviation

In Words

1. Find the mean of the sample data set.
2. Find deviation of each entry.
3. Square each deviation.
4. Add to get the sum of squares.

In Symbols

$$\bar{x} = \frac{\Sigma x}{n}$$

$$x - \bar{x}$$

$$(x - \bar{x})^2$$

$$SS_x = \Sigma(x - \bar{x})^2$$

Finding the Sample Variance & Standard Deviation

In Words

5. Divide by $n - 1$ to get the **sample variance**.
6. Find the square root to get the **sample standard deviation**.

In Symbols

$$s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1}$$

$$s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}$$

Example: Finding Sample Variance & Standard Deviation

- In a study of high school football players who sustained concussions, researchers placed the players in two groups. Players that recovered from their concussions in 14 days or less were placed in Group 1. Those that took more than 14 days were placed in Group 2. The recovery times (in days) for Group 1 are listed below. Find the sample variance and standard deviation of the recovery times. (*Adapted from The American Journal of Sports Medicine*)

8 10 4 6 7 7 9 10 7 6 5 11

Solution: Finding Sample Variance & Standard Deviation

- Find $\sum x$.
- Find the standard deviation for each data entry, s .
- Find the sum of the squares, SS_x .

Time x	Deviation $x - \bar{x}$	Squares $(x - \bar{x})^2$
8	0.5	0.25
10	2.5	6.25
4	-3.5	12.25
6	-1.5	2.25
7	-0.5	0.25
7	-0.5	0.25
9	1.5	2.25
10	2.5	6.25
7	-0.5	0.25
6	-1.5	2.25
5	-2.5	6.25
11	3.5	12.25
$\sum x = 90$		$SS_x = 51$

Solution: Finding Sample Variance & Standard Deviation

For this data set, $n = 12$ and $\sum x = 90$. The mean is $\bar{x} = 90/12 = 7.5$. To calculate s^2 and s , note that $n - 1 = 12 - 1 = 11$. $SS_x = 51$.

- **Sample Variance**

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{51}{11} \approx 4.6$$

- **Sample Standard Deviation**

$$s = \sqrt{\frac{51}{11}} \approx 2.2$$

The sample variance is about 4.6, and the sample standard deviation is about 2.2 days.

Example: Using Technology to Find the Standard Deviation

Sample office rental rates (in dollars per square foot per year) for Los Angeles are shown in the table. Use technology to find the mean rental rate and the sample standard deviation.
(Adapted from LoopNet.com)

Office rental rates		
42	14	46
30	23	50
26	39	13
48	42	24
30	35	36
45	23	72
35	34	30
60	30	64
14	36	23
42	35	72

Solution: Using Technology to Find the Standard Deviation

MINITAB

Descriptive Statistics: Rental Rates

Variable	N	Mean	SE Mean	StDev	Minimum
Rental Rates	30	37.10	2.83	15.48	13.00
Variable	Q1	Median	Q3	Maximum	
Rental Rates	26.00	35.00	45.00	72.00	

EXCEL

	A	B
1	Mean	37.1
2	Standard Error	2.826943
3	Median	35
4	Mode	30
5	Standard Deviation	15.48381
6	Sample Variance	239.7483
7	Kurtosis	0.2787
8	Skewness	0.689938
9	Range	59
10	Minimum	13
11	Maximum	72
12	Sum	1113
13	Count	30

TI-84 PLUS

1-Var Stats
 $\bar{x}=37.1$
 $\Sigma x=1113$
 $\Sigma x^2=48245$
 $Sx=15.48380689$
 $\sigma x=15.22355631$
 $\downarrow n=30$

Sample Mean

Sample Standard Deviation

From the displays,
 you can see that
 $\bar{x} \approx 37.1$ and
 $s \approx 15.48$.

Exercise 1

Find the range, mean, variance, and standard deviation of the population of a data set.

Drunk Driving The number of alcohol-impaired crash fatalities (in thousands) per year from 2010 through 2019.

10.1 9.9 10.3 10.1 9.9 10.3 11.0 10.9 10.7 10.1

Solution 1

x	$x - \mu$	$(x - \mu)^2$

Exercise 2

Find the range, mean, variance, and standard deviation of the sample of a data set.

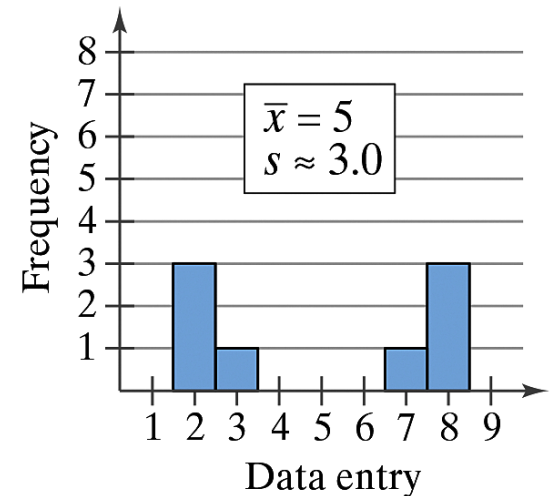
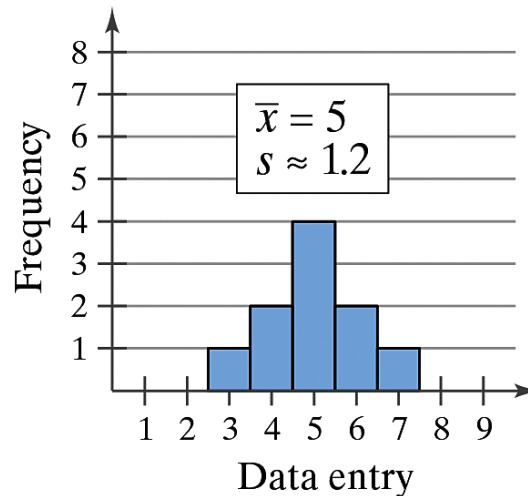
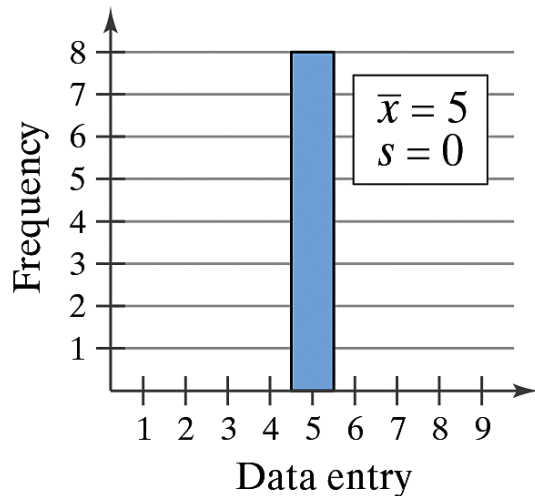
Pregnancy Durations The durations (in days) of pregnancies for a random sample of pregnant people.

277	291	295	280	268	278	291
277	282	279	296	285	269	293
267	281	286	269	264	299	275

Solution 2

Interpreting Standard Deviation

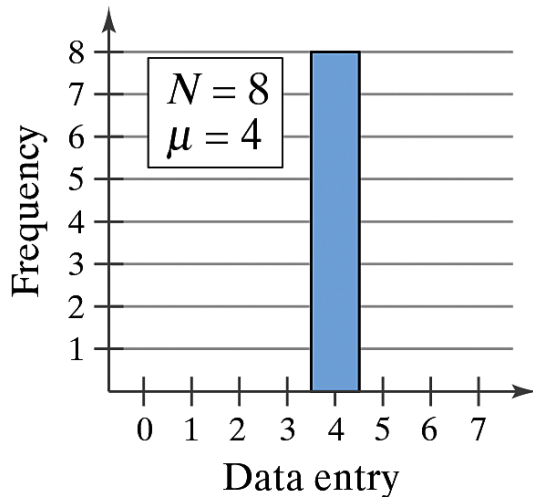
- Standard deviation is a measure of the typical amount an entry deviates from the mean.
- The more the entries are spread out, the greater the standard deviation.



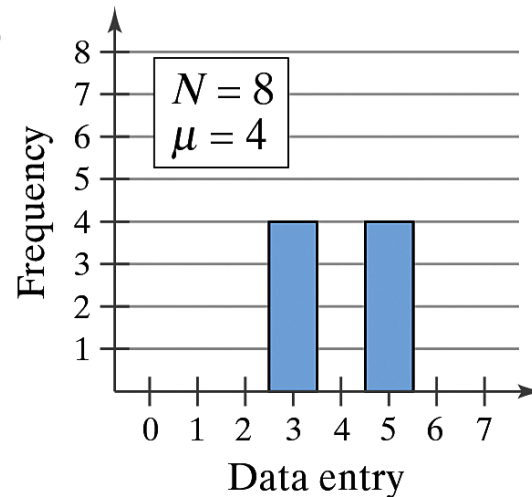
Example: Estimating Standard Deviation

Without calculating, estimate the population standard deviation of each data set.

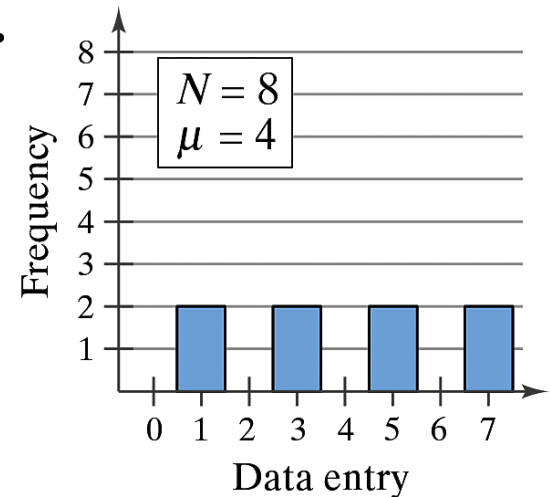
1.



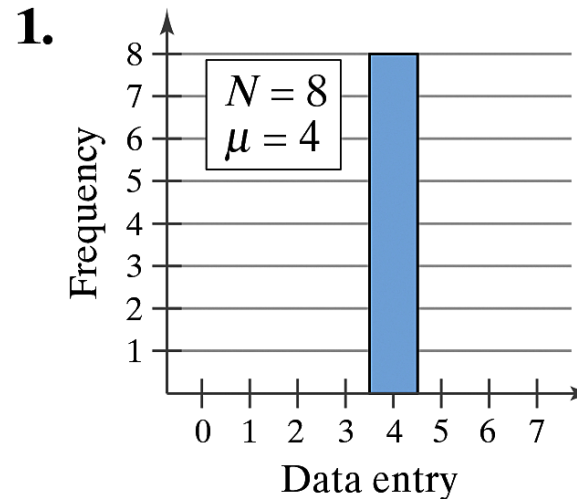
2.



3.

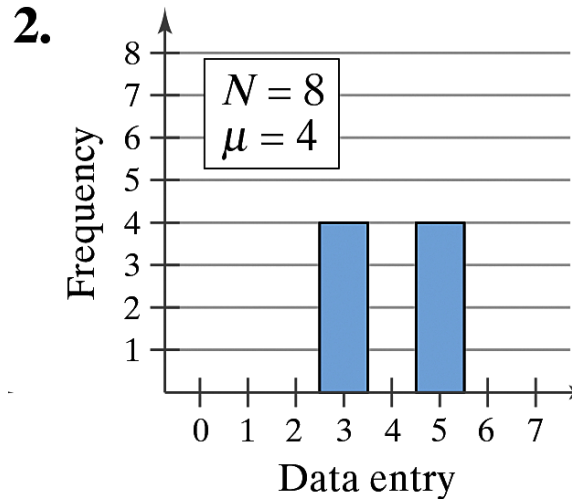


Solution: Estimating Standard Deviation



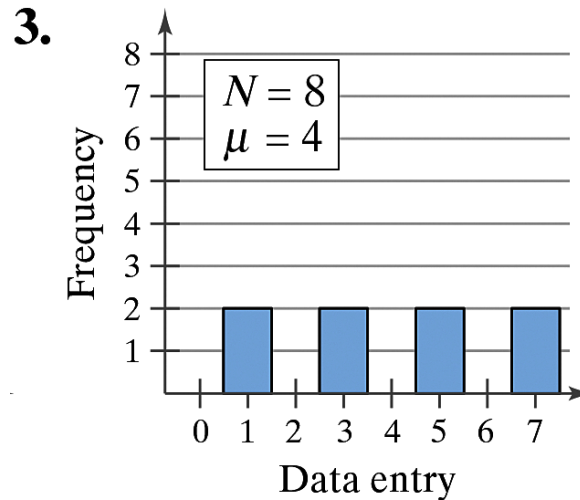
1. Each of the eight entries is 4. The deviation of each entry is 0, so $\sigma = 0$.

Solution: Estimating Standard Deviation



2. Each of the eight entries has a deviation of ± 1 . So, the population standard deviation should be 1. By calculating, you can see that $\sigma = 1$.

Solution: Estimating Standard Deviation



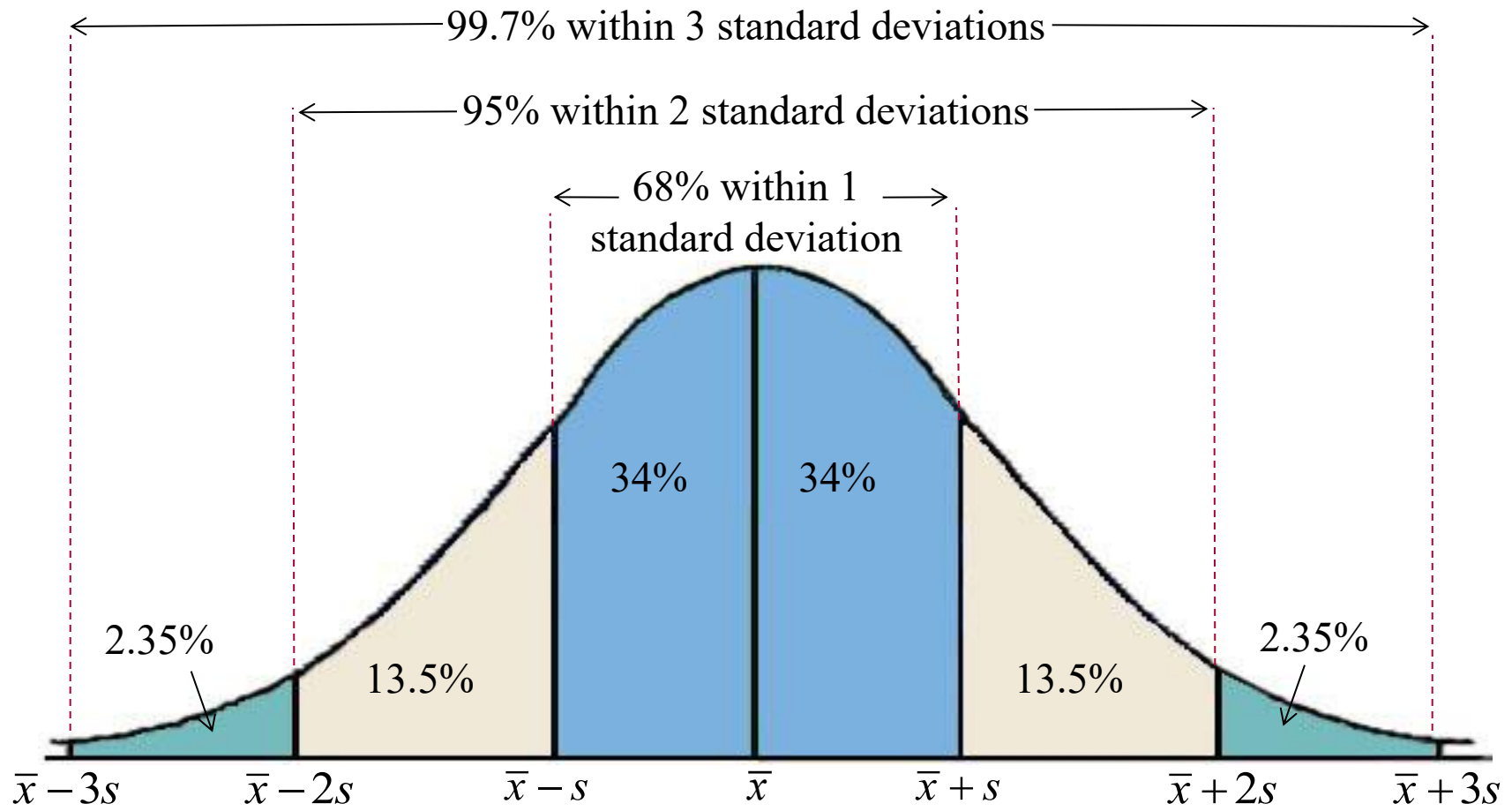
3. Each of the eight entries has a deviation of ± 1 or ± 3 . So, the population standard deviation would be about 2. By calculating, you can see that σ is greater than 2, with $\sigma \approx 2.2$.

Interpreting Standard Deviation: Empirical Rule (68 – 95 – 99.7 Rule)

For data with a (symmetric) bell-shaped distribution, the standard deviation has the following characteristics:

- About **68%** of the data lie within one standard deviation of the mean.
- About **95%** of the data lie within two standard deviations of the mean.
- About **99.7%** of the data lie within three standard deviations of the mean.

Interpreting Standard Deviation: Empirical Rule (68 – 95 – 99.7 Rule)



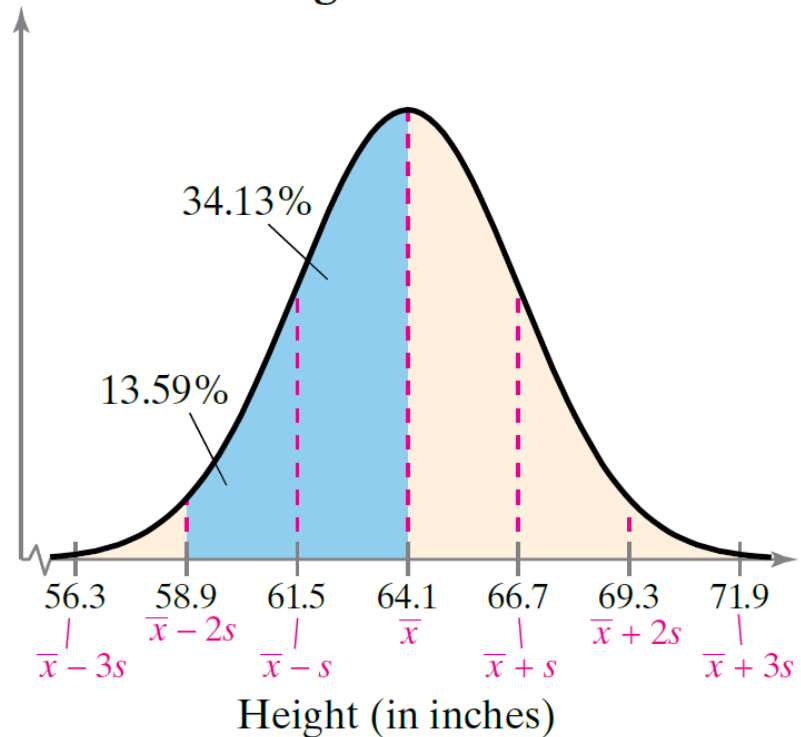
Example: Using the Empirical Rule

In a survey conducted by the National Center for Health Statistics, the sample mean height of women in the United States (ages 20-29) was 64.1 inches, with a sample standard deviation of 2.6 inches. Estimate the percent of the women whose heights are between 58.9 inches and 64.1 inches. *(Adapted from National Center for Health Statistics)*

Solution: Using the Empirical Rule

- The distribution of women's heights is shown. Because the distribution is bell-shaped, you can use the Empirical Rule.
- The mean height is 64.1 inches.

**Heights of Women in the U.S.
Ages 20–29**



Solution: Using the Empirical Rule

- When you subtract two standard deviations from the mean height, you get

$$\bar{x} - 2s = 64.1 - 2(2.6) = 58.9.$$

- Because 58.9 is two standard deviations below the mean height, the percent of the heights between 58.9 and 64.1 inches is about $13.59\% + 34.13\% = 47.72\%$.

So, about 47.72% of women are between 58.9 and 64.1 inches tall.

Exercise 3

The mean value of land and buildings per acre from a sample of farms is \$1500, with a standard deviation of \$200. The data set has a bell-shaped distribution. Estimate the percent of farms whose land and building values per acre are between \$1300 and \$1700.

Solution 3

Solution 3

Solution 3

Solution 3

Chebychev's Theorem

- The portion of any data set lying within k standard deviations ($k > 1$) of the mean is at least:

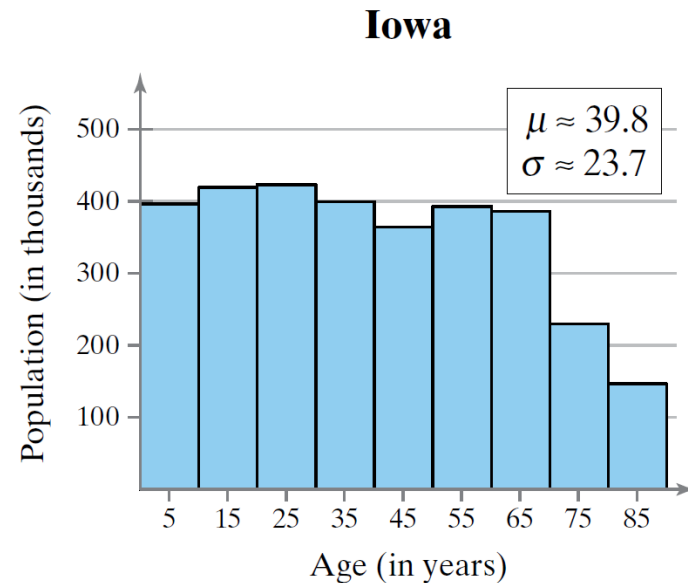
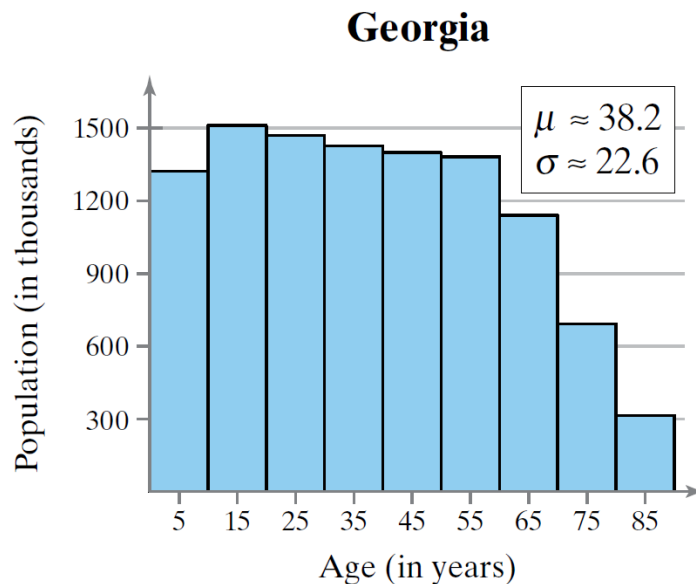
$$1 - \frac{1}{k^2}$$

- $k = 2$: In any data set, at least $1 - \frac{1}{2^2} = \frac{3}{4}$, or 75% of the data lie within 2 standard deviations of the mean.
- $k = 3$: In any data set, at least $1 - \frac{1}{3^2} = \frac{8}{9}$, or about 88.9% of the data lie within 3 standard deviations of the mean.

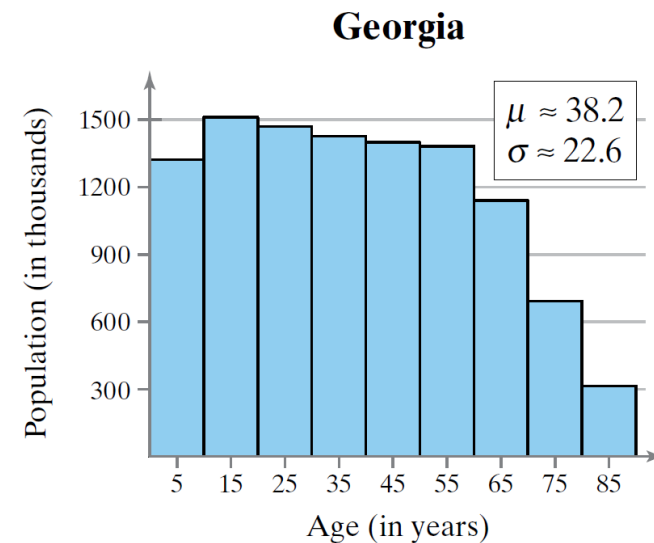
Example: Using Chebychev's Theorem

The age distributions for Georgia and Iowa are shown in the histograms. Apply Chebychev's Theorem to the data for Georgia using $k = 2$. What can you conclude? Is an age of 90 unusual for a Georgia resident? Explain.

(Source: Based on U.S. Census Bureau)



Solution: Using Chebychev's Theorem



$k = 2$: $\mu - 2\sigma = 38.2 - 2(22.6) = -7.0$ (use 0 since age can't be negative)

$$\mu + 2\sigma \approx 38.2 + 2(22.6) = 83.4$$

You can say that at least 75% of the population of Georgia is between 0 and 83.4 years old. Also, an age of 90 lies more than two standard deviations from the mean. So, this age is unusual.

Exercise 4

The mean time in a woman's 400-meter dash is 57.07 seconds with a standard deviation of 1.05. Apply Chebychev's Theorem to the data using $k = 2$. Interpret the results.

Solution 4

Solution 4

Solution 4

Solution 4

Standard Deviation for Grouped Data

Sample standard deviation for a frequency distribution

- $s = \sqrt{\frac{\Sigma(x - \bar{x})^2 f}{n - 1}}$ where $n = \Sigma f$ (the number of entries in the data set)
- When a frequency distribution has classes, estimate the sample mean and standard deviation by using the midpoint of each class.

Example: Finding the Standard Deviation for Grouped Data

You collect a random sample of the number of children per household in a region. Find the sample mean and the sample standard deviation of the data set.

Number of Children in 50 Households				
1	3	1	1	1
1	2	2	1	0
1	1	0	0	0
1	5	0	3	6
3	0	3	1	1
1	1	6	0	1
3	6	6	1	2
2	3	0	1	1
4	1	1	2	2
0	3	0	2	4

Solution: Finding the Standard Deviation for Grouped Data

- First construct a frequency distribution.

- Find the mean of the frequency distribution.

$$\bar{x} = \frac{\Sigma xf}{n} = \frac{91}{50} \approx 1.8$$

The sample mean is about 1.8 children.

x	f	xf
0	10	$0(10) = 0$
1	19	$1(19) = 19$
2	7	$2(7) = 14$
3	7	$3(7) = 21$
4	2	$4(2) = 8$
5	1	$5(1) = 5$
6	4	$6(4) = 24$

$$\Sigma f = 50 \quad \Sigma(xf) = 91$$

Solution: Finding the Standard Deviation for Grouped Data

- Determine the sum of squares.

x	f	$x - \bar{x}$	$(x - \bar{x})^2$	$(x - \bar{x})^2 f$
0	10	$0 - 1.82 = -1.82$	$(-1.82)^2 = 3.3124$	$3.31(10) = 33.124$
1	19	$1 - 1.82 = -0.82$	$(-0.82)^2 = 0.6724$	$0.67(19) = 12.7756$
2	7	$2 - 1.82 = 0.18$	$(0.18)^2 = 0.0324$	$0.04(7) = 0.2268$
3	7	$3 - 1.82 = 1.18$	$(1.18)^2 = 1.3924$	$1.39(7) = 9.7468$
4	2	$4 - 1.82 = 2.18$	$(2.18)^2 = 4.7524$	$4.75(2) = 9.5048$
5	1	$5 - 1.82 = 3.18$	$(3.18)^2 = 10.1124$	$10.11(1) = 10.1124$
6	4	$6 - 1.82 = 4.18$	$(4.18)^2 = 17.4724$	$17.47(4) = 69.8896$

$$\Sigma(x - \bar{x})^2 f = 145.38$$

Solution: Finding the Standard Deviation for Grouped Data

- Find the sample standard deviation.

$$s = \sqrt{\frac{\Sigma(x - \bar{x})^2 f}{n - 1}} = \sqrt{\frac{145.38}{49}} \approx 1.7$$

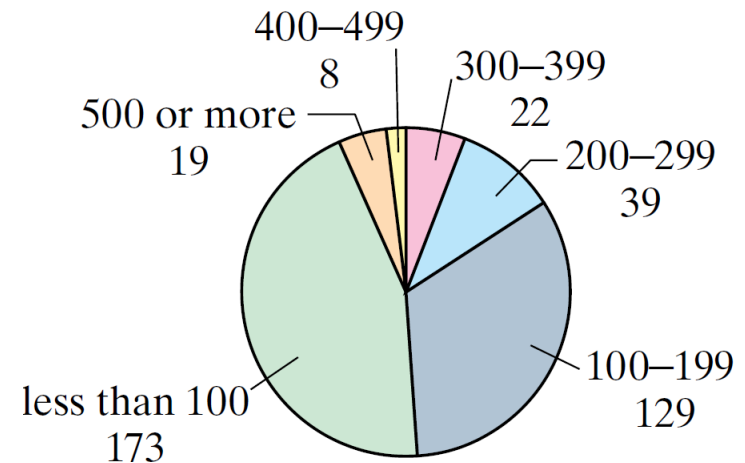
The standard deviation is about 1.7 children.

Example: Using Midpoints of Classes

The figure shows the price ranges (in thousands of dollars) and corresponding number of homes recently listed for sale in a mid-sized U.S. city. Make a frequency distribution for the data. Use the table to estimate the sample mean and the sample standard deviation of the data set. (*Adapted from National Association of Realtors and Move, Inc.*)

Sales Prices of Homes

(in thousands of dollars) in a mid-sized city



Solution: Using Midpoints of Classes

Begin by using a frequency distribution to organize the data. Because the class of \$500 thousand or more is open-ended, you must choose a value to represent the midpoint, such as 599.5 thousand.

Solution: Using Midpoints of Classes

Class	x	f	xf
0 – 99	49.5	173	8563.5
100 – 199	149.5	129	19,285.5
200 – 299	249.5	39	9730.5
300 – 399	349.5	22	7689.0
400 – 499	449.5	8	3596.0
500 +	599.5	19	11,390.5
		$\Sigma = 390$	$\Sigma = 60,255.0$

$$\bar{x} = \frac{\Sigma xf}{n} = \frac{60,255.0}{390} = 154.5 \text{ is the sample mean}$$

Solution: Using Midpoints of Classes

$x - \bar{x}$	$(x - \bar{x})^2$	$(x - \bar{x})^2 f$
-105	11,025	1,907,325
- 5	25	3225
95	9025	351,975
195	38,025	836,550
295	87,025	696,200
445	198,025	3,762,475
		$\Sigma = 7,557,750$

$$s = \sqrt{\frac{\sqrt{(x - \bar{x})^2 f}}{n - 1}} = \sqrt{\frac{7,557,750}{389}} \approx 139.4$$

is the sample standard deviation

Solution: Using Midpoints of Classes

- An estimate for the sample mean is \$154,500 per year, and an estimate for the sample standard deviation is \$139,400 per year.

Coefficient of Variation

Coefficient of Variation (CV)

- Describes the standard deviation of a data set as a percent of the mean.
- Population data set:

$$CV = \frac{\sigma}{\mu} \cdot 100\%$$

- Sample data set:

$$CV = \frac{s}{\bar{x}} \cdot 100\%$$

Example: Comparing Variation in Different Data Sets

The table shows the population heights (in inches) and weights (in pounds) of the members of a basketball team. Find the coefficient of variation for the heights and the weights. Then compare the results.

Heights and Weights of a Basketball Team

Heights	72	74	68	76	74	69	72	79	70	69	77	73
Weights	180	168	225	201	189	192	197	162	174	171	185	210

Solution: Comparing Variation in Different Data Sets

The mean height is $\mu \approx 72.8$ inches with a standard deviation of $\sigma \approx 3.3$ inches. The coefficient of variation for the heights is

$$CV_{\text{height}} = \frac{\sigma}{\mu} \cdot 100\% = \frac{3.3}{72.8} \cdot 100\% \approx 4.5\%$$

Solution: Comparing Variation in Different Data Sets

The mean weight is $\mu \approx 187.8$ pounds with a standard deviation of $\sigma \approx 17.7$ pounds. The coefficient of variation for the weights is

$$CV_{\text{weight}} = \frac{\sigma}{\mu} \cdot 100\% = \frac{17.7}{187.8} \cdot 100\% \approx 9.4\%$$

The weights (9.4%) are more variable than the heights (4.5%).